

Valgrind Memory Report - NanoServe C Engine

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****Date:**** 2026-07-25
****Tool:**** Valgrind 3.18.1 Memcheck
****Target:**** `tests/valgrind\_engine` (200 init/infer/cleanup cycles)
****Verdict:**** ZERO DEFINITE LEAKS

Executive summary

The C++ inference engine was exercised under Valgrind Memcheck for ****200 complete lifecycles**** (create engine -> infer -

Command

./scripts/valgrind.sh
Equivalent:
valgrind --leak-check=full --error-exitcode=1 \
./tests/valgrind_engine

Heap summary (excerpt)

OK 200 cycles
HEAP SUMMARY:
in use at exit: 0 bytes in 0 blocks
total heap usage: 6,207 allocs, 6,207 frees, 3,368,984,080 bytes allocated
All heap blocks were freed -- no leaks are possible
ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
Full log: `documentation/valgrind\_report.txt`

What was tested

Python / dlopen note

An earlier Valgrind run on the Python loader reported ****192 bytes**** still reachable from `dlopen` of the engine `.so`. T

Prior requirements

sudo apt-get install -y valgrind build-essential
./install.sh
./scripts/valgrind.sh

Fixes validated by this report

- Null checks on failed `engine\_create`
- Pool leak prevention on partial initialization failure
- Safe destroy when handles are null
- CUDA malloc failure paths (when CUDA enabled)

Conclusion

The NanoServe C++ engine passes Valgrind Memcheck with ****0 bytes definitely lost**** and ****0 error contexts****. Safe for lo